

SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project

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IBM Data Science Professional Certificate – Coursera

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GitHub Repository:

<https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone>

Executive Summary

- **Objective:** Predict whether the Falcon 9 first stage will land successfully using historical launch and mission characteristics. The prediction can support launch-cost estimation because successful first-stage recovery enables rocket reuse.
- **Data and Methodology:** Launch data were collected from the SpaceX REST API and Wikipedia using the completed API Data Collection and Web Scraping notebooks. The cleaned dataset was analyzed using Python visualizations and ten SQL queries. Detailed collection, wrangling, EDA, and SQL results are presented on Slides 4–16.
- **Predictive Modeling:** Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors models were trained and tuned using cross-validation.
- **Key Findings:** Exploratory analysis suggests that landing outcomes are associated with flight progression, payload mass, launch site, and orbit type. All four tuned models achieved approximately 83.3% test accuracy in the current experiment.
- **Business Value:** A reliable landing-success model can help estimate mission risk and the potential economic benefit of reusable launch systems.

Introduction

- **Project Background:** SpaceX advertises Falcon 9 launches at approximately USD 62 million, which is substantially lower than many traditional launch alternatives. A major source of cost savings is the ability to recover and reuse the first stage.
- **Business Problem:** The cost of a launch depends partly on whether the first stage can be recovered. Predicting the landing outcome can therefore support launch-cost and mission-risk estimation.
- **Data Science Question:** Can launch characteristics such as payload mass, launch site, orbit type, booster information, and flight history be used to predict whether the Falcon 9 first stage will land successfully?
- **Target Variable**
Class = 1: successful first-stage landing
Class = 0: unsuccessful or non-successful landing outcome
- **Completed Project Deliverables:**
 - ✓ SpaceX API data collection — Slide 4
 - ✓ Wikipedia web scraping — Slide 5
 - ✓ Data wrangling — Slide 6
 - ✓ EDA visualizations — Slides 7–9
 - ✓ SQL analysis — Slides 10–16
 - ✓ Folium map — Slides 17–18 and 20
 - ✓ Plotly Dash application — Slides 19 and 21
 - ✓ Classification modeling — Slides 22–23

Data Collection – SpaceX API

- **Data Source**

Launch information was retrieved from the SpaceX REST API using `requests.get()`.

- **Processing Steps**

1. Retrieved launch records in JSON format.
2. Used `pandas.json_normalize()` to flatten nested JSON objects.
3. Extracted relevant variables, including flight number, launch date, payload mass, orbit, launch site, booster information, and landing outcome.
4. Filtered the records to retain Falcon 9 launches only.
5. Replaced missing payload-mass values with the mean of the observed payload

masses.

GitHub URL for API Notebook:

<https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection – Web Scraping

Data Source

The notebook defines a function for reading Falcon 9 and Falcon Heavy launch tables from Wikipedia using `pandas.read_html()`.

For the saved execution, live web scraping was disabled and the IBM Skills Network course snapshot was used.

Processing Steps

1. Read and flattened the launch-table headers.
2. Detected tables containing flight and launch information.
3. Used the IBM course snapshot when live scraping was disabled or unavailable.
4. Standardized column names and whitespace.
5. Saved 101 launch records to `spacex_web_scraped.csv`.

GitHub URL for Web Scraping Notebook:

<https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb>

Data Wrangling and Feature Preparation

Data Cleaning

- Inspected missing values and data types.
- Replaced missing payload-mass values using the mean of the available observations.
- Standardized column formats and retained Falcon 9 launch records.

Target Creation

- Examined the original landing-outcome categories.
- Mapped successful landing outcomes to Class = 1.
- Mapped unsuccessful or non-successful outcomes to Class = 0.

Modeling Dataset

- Selected relevant launch, payload, orbit, site, booster, and landing-pad variables.
- Removed fields that were not appropriate for prediction or were direct identifiers.

GitHub URL for Data Wrangling Notebook:

<https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb>

EDA with Data Visualization

Exploratory Questions

- Does landing success become more frequent as the flight number increases?
- How does payload mass relate to landing outcome?
- Do landing-success rates differ by launch site and orbit type?
- How has landing success changed over time?

Visualizations

- Scatter plots: flight number and payload mass versus landing outcome.
- Site-level plots: landing success across launch sites.
- Bar charts: success rate and launch count by orbit type.
- Line charts: annual landing-success trend.

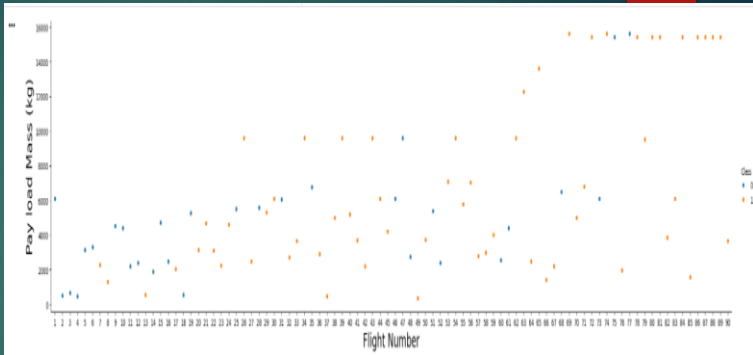
Tools

Pandas, Matplotlib, and Seaborn.

GitHub URL for EDA Visualization Notebook:

https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/eda_visualization.ipynb

Landing Outcome by Flight Number and Payload Mass

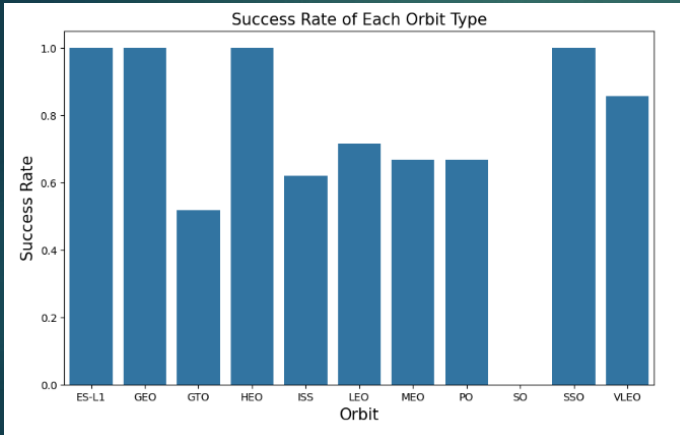


Key Findings:

- Successful landings become more frequent among later Falcon 9 flights, which is consistent with operational improvement over time.
- Payload mass alone does not perfectly separate successful and failed landings.
- Landing outcome should therefore be analyzed together with launch site, orbit type, and flight history.

Interpretation Note: These relationships are observational and should not be interpreted as proof of causality.

Landing Success Rate by Orbit Type



Key Findings:

- Landing-success rates vary across orbit categories.
- Some orbit types display very high success rates, but their launch counts may be small.
- Both the success percentage and the number of launches must be considered before comparing orbit performance.

Recommended Enhancement: Display the number of launches above each bar or add a companion count chart.

EDA with SQL

SQL Analysis Environment

The launch dataset was loaded into a SQLite database and analyzed through ten completed SQL tasks in a Jupyter Notebook. The queries used SELECT DISTINCT, LIKE, BETWEEN, aggregate functions, grouping, sorting, date filtering, and subqueries. Query code and outputs are presented on Slides 11–16.

Analysis Objectives

- Identify the unique launch sites.
- Calculate total and average payload masses.
- Analyze mission and landing outcomes.
- Identify the first successful ground-pad landing.
- Compare booster versions within selected payload ranges.
- Analyze landing outcomes across time periods.

SQL Techniques: SELECT DISTINCT, WHERE, LIKE, BETWEEN, GROUP BY, ORDER BY, COUNT, SUM, AVG, MIN, and subqueries.

Completed SQL Notebook: GitHub URL for SQL Notebook: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/jupyter_labs_eda_sql_coursera_sqlite.ipynb

EDA with SQL Results

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

EDA with SQL Results

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

Done.

Total_Payload_Mass

45596

EDA with SQL Results

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Average_Payload_Mass
```

```
2928.4
```

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql SELECT MIN(Date) AS First_Successful_Landing_Date FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First_Successful_Landing_Date
```

```
2015-12-22
```

EDA with SQL Results

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) as Total_Number FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total_Number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA with SQL Results

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

EDA with SQL Results

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Month Landing_Outcome Booster_Version Launch_Site
```

```
01 Failure (drone ship) F9 v1.1 B1012 CCAFS LC-40
```

```
04 Failure (drone ship) F9 v1.1 B1015 CCAFS LC-40
```

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql SELECT Landing_Outcome, COUNT(*) as Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Landing_Outcome Outcome_Count
```

```
No attempt 10
```

```
Success (drone ship) 5
```

```
Failure (drone ship) 5
```

```
Success (ground pad) 3
```

```
Controlled (ocean) 3
```

```
Uncontrolled (ocean) 2
```

```
Failure (parachute) 2
```

```
Precluded (drone ship) 1
```


Folium Methodology

Folium Geospatial Analysis

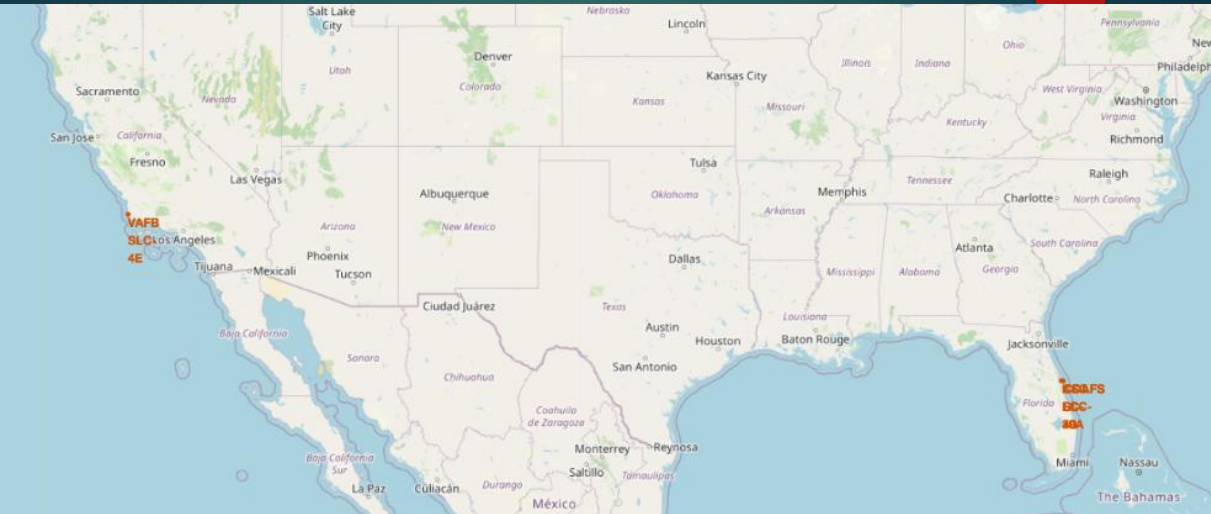
- Plotted all launch-site locations on an interactive map.
- Added labels for each launch site.
- Used color-coded markers to distinguish successful and unsuccessful landings.
- Applied MarkerCluster to organize overlapping launch markers.
- Measured distances from launch sites to coastlines, roads, railways, and nearby cities.

Purpose

Examine whether launch sites share common geographic and infrastructure characteristics.

GitHub URLs: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Folium Map Results



Plotly Dash Methodology

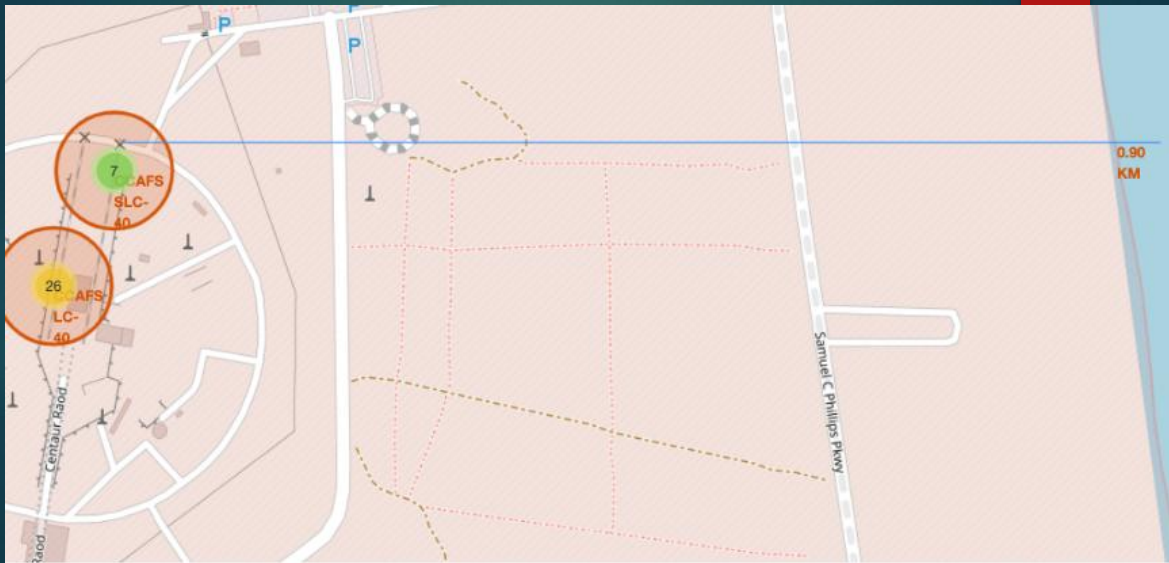
Plotly Dash Application

- Added a launch-site dropdown selector.
- Added a payload-mass range slider.
- Created a pie chart showing landing outcomes by launch site.
- Created a scatter plot showing payload mass versus landing outcome.
- Used booster-version categories to compare performance across mission configurations.

GitHub URL for Plotly Dash Application:

https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_dash_app.py

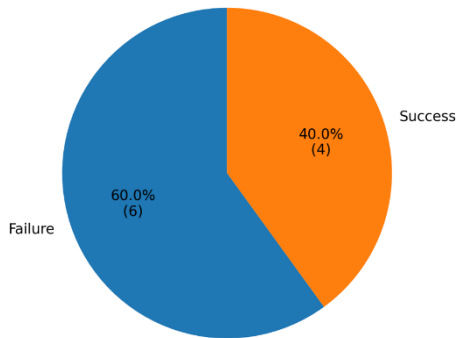
Folium Map Results



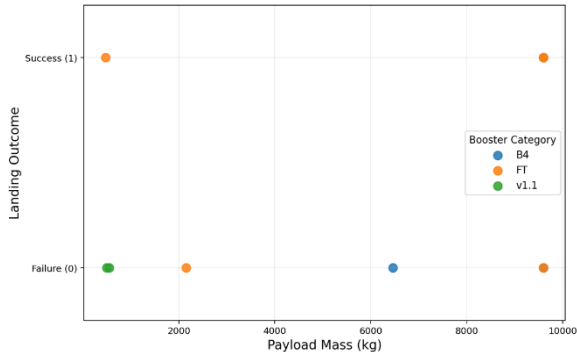
Plotly Dash Results

21/1
4

Landing Outcomes — VAFB SLC-4E
10 launches



Payload Mass vs. Landing Outcome — VAFB SLC-4E



Predictive Analysis

Model Selection

Because test accuracy was tied, no model can be described as uniquely superior based on test accuracy alone. The final model was selected primarily using the meancross-validation accuracy. Test-set accuracy and the confusionmatrix were used as secondary evaluation criteria.

Error Analysis

The confusion matrix shows how many successful and failed landings were classified correctly and incorrectly. False-success predictions are especially important because they may cause launch risk to be underestimated.

Predictive Analysis

- Conclusion Payload mass, launch site, orbit type, and flight number are important indicators of Falcon 9 first-stage landing success.
- Logistic Regression, SVM, Decision Tree, and KNN achieved approximately 83.3% test accuracy. Decision Tree or SVM may be selected based on the tuning results.
- The predictive models can support launch-cost estimation by helping determine whether the first stage is likely to be recovered.
- Overall, the analysis demonstrates how data science can provide useful insights for companies competing in the commercial launch industry.

Conclusion

1. Exploratory Findings

Landing success becomes more frequent among later Falcon 9 flights. Landing outcomes also vary across payload ranges, launch sites, and orbit types, although these relationships should be interpreted as associations rather than causal effects.

2. SQL Findings

The dataset contains four unique launch sites. The average payload mass for F9 v1.1 missions is 2,928.4 kg, and the first successful ground-pad landing in the dataset occurred on December 22, 2015.

3. Predictive Modeling

Logistic Regression, SVM, Decision Tree, and KNN each achieved approximately 83.3% test accuracy after tuning. Since their test accuracies were tied, cross-validation results and the error profile should be used to select the final model.

4. Business Implication

Landing-success predictions can support preliminary mission-risk and launch-cost estimation by indicating the likelihood that the first stage can be recovered and reused.

5. Limitations and Future Work

The analysis is based on a relatively small historical dataset and may be affected by class imbalance and changes in launch technology over time. Future work should include newer launch records, additional operational variables, and probability-based model calibration.